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# GAN-CNN Ensemble: A Robust Deepfake Detection Model of Social Media Images Using Minimized Catastrophic Forgetting and Generative Replay Technique

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## Abstract

Deep-fake photographs are difficult to discern from real ones, especially when utilized in social media platforms. Anyone can willfully create disinformation about public personalities, politicians, and celebrities using these deep fake photographs. So, it is an important need of society to work for an effective model for its detection. The models for deep fake detection commonly use CNN-based detectors. These detectors experience a drop in performance when used for transfer learning or continual learning techniques. A significant limitation in this process is CNN's catastrophe forgetting defect. For the solution of this problem, a Generative replay technique in the form of a GAN-CNN model is implemented that works to minimize this catastrophe forgetting issue that further helps for better detection. It involves generating and storing samples from previous tasks and then replaying them during the training of new tasks which makes CNN more robust to identify deep fakes. The GAN model used in this work is traditional DCGAN improved with necessary adjustments to achieve training stability. It is observed that the model attained a good accuracy of 98.67%(training) ,70.08% (testing) and minimum loss with a value of 0.0337 for 100 epochs. Also, it acquired good precision values of 68% and 72%, Recall values are 74% and 66%, and F1 scores of 71% and 69% for classes 0 and 1 respectively. The model outcome is found stable and reliable in deep fake detection under dynamic training conditions. Optimum values of evaluation parameters ensure the model's capacity to learn new tasks preserving the existing task-learning knowledge.

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**Keywords.** Deep Learning; CNN;GAN;Catastrophe forgetting;CNN continual learning;Lifelong learning; GAN-CNN deep fake detector.

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## 1. Introduction

Generative Adversarial Networks (GANs) are used to create realistic visual output from a probability distribution.

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The machine learning research community's involvement in GANs is expanding [1]. Generative adversarial networks (GANs) are neural network-based models that use supervised learning and synthetic data to achieve unsupervised learning without predetermined results [2]. They address the shortcomings of other generative models by introducing adversarial learning. GANs aim to imply data structures instead of maximizing likelihood [3]. In GAN, the discriminator and generator compete to create realistic samples. GAN is extensively used in deepfake techniques, which can have negative effects if used unethically. Deep fake images, difficult to distinguish from real ones, can create false information about public figures like politicians, and celebrities, including bribes, prejudice, and anti-national radical gatherings [4]. Digital forensic experts are developing deepfake image detection methods to confirm the reliability of digital images, using CNN models and dynamic learning environments to identify and separate manipulated images [5].

Catastrophic forgetting is a common issue in CNN models where neural networks trained with traditional backpropagation fail to retain previously learned content during distributed processing changes, hindering their ability to adapt to Continual Learning (CL) or Lifelong Learning situations. The problem of catastrophe forgetting affects neural networks, particularly CNN models used as deepfake detectors. This poses an obstacle to creating generalized artificial learning systems. Continual learning is a characteristic of natural intelligence and a fruitful area of research in machine learning [6]. It has resulted in the development of generative replay, which minimizes catastrophic forgetting by producing pseudo-samples for previous tasks and learning them alongside new tasks. Generative replay, also known as pseudo-rehearsal, is a successful and generic strategy for continual learning shown in Fig. 1.

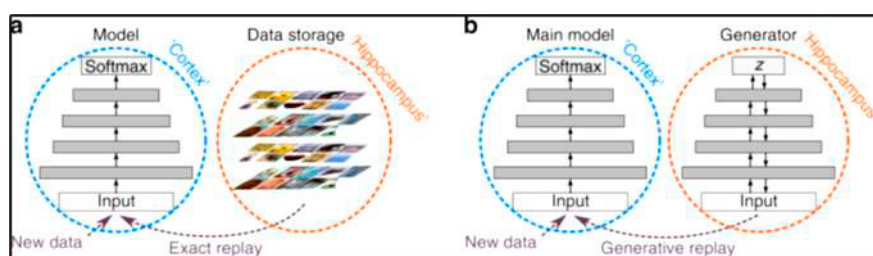


Fig. 1. Model of Brain-Inspired Generative Replay Model of Continual Learning of Neural Network [7]. ((a). Exact or experience replay, which views the hippocampus as a memory buffer in which experiences can simply be stored, akin to traditional views of episodic memory. b. Generative replay with a separate generative model, which views the hippocampus as a generative neural network and replays as a generative process).

This research proposes a GAN-CNN ensemble model to minimize catastrophic forgetting of CNN models and improve Deepfake detection. The model uses adversarial training of the DCGAN model for continual learning, creating and storing samples from previous tasks and replaying them during training of new tasks of the CNN model a generative replay technique. This approach makes CNN adaptive in detecting deep fakes, allowing it to work in a dynamic environment with variable datasets. The model uses a DCGAN model tuned for training stability, and optimal evaluation parameter values enhance its ability to learn new tasks while retaining old task-learning expertise.

#### Contributions-

- An evolutionary GAN-CNN ensemble model significantly enhances deepfake detection accuracy, achieving a 98.67% rate, thereby improving content verification reliability, and identifying manipulated media.
- A robust model that effectively combats catastrophic forgetting in CNN models by utilizing a GAN-based approach, ensuring resilience in adapting to new data and retaining prior knowledge.
- The GAN-CNN ensemble effectively manages time, balancing computational efficiency and learning effectiveness, making it a practical solution for real-time or large-scale deepfake detection scenarios.

The rest of the study in this paper is organized in a manner with Section 2 discussing the related studies. Section 3

highlights the methodology used in deriving the model, including the experimentation details. Results and Discussion are analysed in section 4 using different curves including Accuracy, Precision, Recall, and F1-score plots. Section 5 finally concludes the paper with future findings.

## 2. Related Studies

In this section, the related works on the significance of continual learning-based GAN approaches are reviewed. Furthermore, literature on GAN deep fake detection is also scrutinized to align this work correctly with the scope of integrating continual learning in GAN deep fake models.

Researchers compared generative models for sequential image creation challenges using rehearsal, regularization, generative replay, and fine-tuning mechanisms. They calculated generation quality and memory capacity using two metrics. They evaluated sequential tasks using MNIST, Fashion MNIST, and CIFAR10 [8]. Generative replay outperformed all other continuous learning algorithms, and the original GAN performed best among all models. The Lifelong GAN [9] is a novel model that uses knowledge distillation to transfer learned information from older networks to new ones, enabling it to complete image-conditioned generating tasks in a lifelong learning context. The model was validated for both image-conditioned and label-conditioned generation tasks, resulting in both qualitative and quantitative results. A continuous learning technique for generative adversarial networks (GANs) was developed using parameter-efficient feature map modifications and task-specific parameters [10]. The model's performance was enhanced by incorporating residual bias and using Fisher information matrix-based task similarity for efficient continual data generation. [11] Researcher rectification methods for GAN training difficulties and evaluated their theoretical contributions with an emphasis on image/visual applications. Deepfake detection is a fine-grained classification challenge, and a new multi-attentional deepfake discovery network was proposed [12]. The model format includes multiple spatial attention heads, a textural feature boosting block, and an accumulation of low-level textural and high-level semantic features directed by attention maps. This approach aims to focus on local regions and detect subtle artifacts. A deepfake efficacy model was created using two GAN networks and a Cycle-GAN network [13]. The loss was split into two parts and a flow loss was added. The model was used for object transformation and style transfer, resulting in pleasing photos. [14] used Continual Representation utilizing the Distillation (CoReD) technique to improve data efficiency in neural networks. CoReD outperforms state-of-the-art methods in detecting low-quality deepfake videos and GAN-generated images from multiple datasets. [15] Proposes a continuous deep fake detection benchmark (CDDDB) for perpetual deepfake detection, utilizing various models, suitable measurements, and multiclass incremental learning techniques, which are commonly used in continual visual recognition, to address the problem.

## 3. Methodology

This section provides the conceptual framework for the continuous learning-based GAN model on deepfake detection, outlining each element used to build the system designed to achieve the study's objective. Corrections for two key problems were prioritized. (a) catastrophic forgetting of pixel-based neural net models; and (b) a straightforward continuous learning strategy for a deep fake detection model based on GAN that yields accurate results. The GAN-CNN model is a combination of the generating power of the GAN model and the detection power of the CNN model. It generates synthetic samples of previously viewed tasks, which are then combined with the data from the current task to form a combined dataset for CNN model training. The GAN-CNN ensemble model is used as a Generative Replay technique in continuous learning, mitigating catastrophic forgetting by storing and replaying samples from past tasks during task training. This process preserves prior task knowledge, prevents forgetting, and refines the model's capabilities for deepfake detection in various scenarios. The model is exposed to the data from the prior task and learns to retain information while adapting to the new task. The GAN model regenerates the original input and anticipates the network's ability to restore any face. The architecture of the proposed GAN-CNN ensemble model is described in Fig. 2 below.

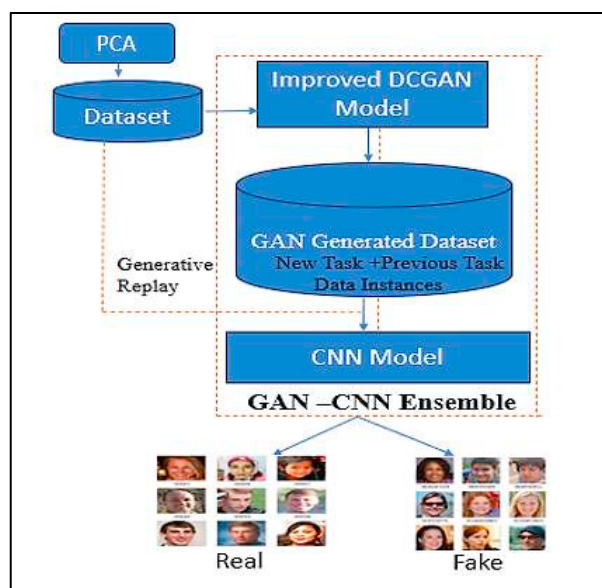


Fig. 2. The architecture of the proposed GAN-CNN ensemble model.

The basic steps involved in the GAN-CNN ensemble model execution are defined below.

**Step 1. Data Collection.** The research methodology begins with data collection, often involving assembling a dataset of social media images, including both real and potentially manipulated images. In our case, we used the “Real and Fake” dataset in a ratio of 70:30 for training and testing purposes.

**Step 2. Preprocessing of Data using PCA.** The data for GAN training undergoes preprocessing using Principal Component Analysis (PCA) to enhance feature extraction by reducing noise and dimensionality and preserving essential information. It converts original features into orthogonal principal components (PC). In our case PC=400 is used as per facial data used. PCs are linear combinations of original features, ordered by variance, with the first component capturing the most data variance.

**Step 3. GAN training.** In the initial phase, the GAN model (DCGAN) is trained using a dataset of real images. The model's GAN component learns to generate synthetic data that closely mimics the original data.

**Step 4. Creating Synthetic Samples.** Once trained, the GAN is used to produce synthetic samples that mirror real-world images and is provided with dimensionality-reduced features derived from PCA. To avoid catastrophic forgetfulness, these synthetic samples are saved for future use.

**Step 5. Generative Replay using DCGAN (Synthetic sample reintroduction).** Before addressing a new assignment, the model applies a Generative Replay approach. Prior tasks are reintroduced with synthetic samples created by the GAN that feature dimensionality-reduced data obtained through PCA preprocessing. These synthetic samples are integrated with the data from the present job, resulting in a consolidated dataset containing both historical and new data.

**Step 6. Training of CNN on New Task.** When the model is given a new task, the synthetic samples from the previous job are combined with the data from the current task to produce a mixed dataset. The CNN model is then trained on this merged dataset ensuring its adaptability and robustness in detecting deepfakes in dynamic environments.

**Step 7. Continual Learning.** As the model is exposed to more tasks, the generative replay process for each prior task is repeated, replaying its synthetic samples throughout the training of new tasks. This continuous exposure to previous tasks aids in the retention of previous task knowledge and prevents catastrophic forgetting.

### 3.1 Training Parameters

Parameters used in the experiment are given below mentioned with the values applied during building the model in Table 1.

**Table 1.** Parameters used in the training of the proposed Model.

| Parameters            | Value                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------|
| Batch Size            | 128 pictures                                                                                      |
| Learning Rate         | 0.05                                                                                              |
| Momentum (Beta)       | 0.5                                                                                               |
| Optimizer             | Adam's Optimizer                                                                                  |
| Epochs                | 100                                                                                               |
| Dropout               | 0.2                                                                                               |
| Batch normalization   | [0,1]                                                                                             |
| Weight Initialisation | Random Values                                                                                     |
| Activation function   | ReLU                                                                                              |
| Evaluation Metrics    | Accuracy, Precision, Recall, F1-Score, Generator Loss, Discriminator Loss, and Computational Time |

The batch size of 128 pictures is chosen to balance training speed with memory constraints, efficiently processing a moderate number of images in each pass. A high learning rate of 0.05 is suggested for quicker convergence, especially for complex tasks and architectures. A moderate momentum of 0.5 balances training stability and momentum term influence on gradient descent, enhancing model efficiency. Adam optimizer is chosen for its effectiveness for training deep models and handling varying learning rates. The model uses 100 epochs to manage computational resources and avoid overfitting. A dropout rate of 0.2 indicates modest regularization to prevent overfitting. Batch normalization allows for experimentation with and without normalization, providing flexibility during training. Weight initialization with random values ensures a reasonable starting point. ReLU is a popular activation function for efficiency and mitigating the vanishing gradient problem. Evaluation metrics are standard and comprehensive, covering accuracy, precision, recall, F1-Score, and loss values, providing a comprehensive assessment of model performance.

### 3.2 GAN Model (DCGAN)

In this research, we have employed DCGAN as the basis model to achieve generative deepfake detection. It is made up of two neural networks, the generator and the discriminator. The generator's goal is to create synthetic data that looks like actual data, while the discriminator's job is to discern between real data and the generator's phony data. These two networks are trained in a competitive manner, with the generator attempting to make realistic samples to fool the discriminator, and the discriminator attempting to improve its ability to distinguish between real and false samples. In its architecture, it consists of convolution layers with no max pooling or completely linked layers. Convolutional stride and transposed convolution are used for down sampling and up sampling. DCGAN employs convolutional and convolutional-transpose layers in its generator and discriminator, respectively. Two loss functions—one for the generator and one for the discriminator—are considered in this approach, which uses two neural networks. The distance between the two estimated values being contrasted with one produced by the generator model and the one based on the real data—is what gives rise to the two loss functions. The cross-entropy-based logic of min-max loss is used to work with the loss function. The generator attempts to minimize it (creating artificial samples that are extremely like the real ones), as the title indicates, while the discriminator seeks to maximize it (by distinguishing between fake and real instances).

$$\min_G \max_D E_{x,y} [\log D(x, G(x))] + E_{x,y} [\log(1 - D(G(x), y))] \quad (1)$$

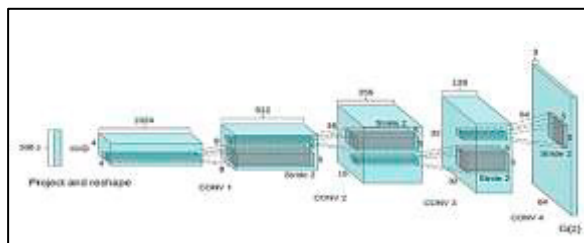


Fig. 3. DC GAN generator's network architecture by Radford et al. [16].

The architecture is modified with necessary adjustments to achieve training stability [17]. The two main functions of GAN with generator and discriminator functions are defined below.

- D uses stride convolutions instead of pooling layers, increasing model expressiveness and trainable parameters, but also increasing the number of trainable parameters.
- G uses transpose convolutions to reduce feature maps and spatial dimensions by projecting sampled noise onto a 2D grid and reshaping it, resulting in a smaller dataset.
- Batch normalization stabilizes preparation and enhances the learning rate. Randomly cropped snapshots are normalized, with center crop and border crops used for the final forecast during the test phase.
- All layers in G except for the output, which generates values in the picture range  $[-1,1]$ , use ReLU activation.
- The discriminator employs leaky ReLU activation across all layers. The model can learn more complicated representations thanks to the nonlinearity of both activation functions.

### 3.2 Dataset

For this study, we also used the Kaggle Real and Fake Face Detection dataset [18], which contains 1,081 real face images and 960 fake face images generated by a Generative Adversarial Network (GAN). The fake face images are generated by training the GAN on the real face images and then using the trained GAN to generate new fake face images as shown in Fig. 4. The experimental data set is divided into a training dataset and a verification dataset. Each experiment was randomly assigned to generate them, corresponding to 70% and 30% respectively. The model is trained on the training dataset and verified on the verification dataset.

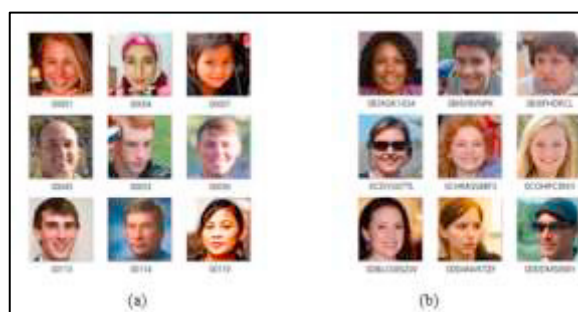


Fig. 4. Description of Real and Fake Images Dataset with (a) showing real faces; and (b) showing fake faces.

### 3.3 Deepfake Detector with Continual Learning Approach (VGG-16)

VGG-16 served as our deepfake detector. Convolution neural net (CNN) architecture as our deepfake detector is employed in the system, which took first place in the 2014 ILSVR (ImageNet) project. It is viewed as one of the top vision model architectures available today. The most remarkable thing about VGG16 is that they are made to give 3x3 filter fully connected layers with a stride 1 but still use the same stride 2 padding and maxpool layer of 2x2 filter [19] rather than creating several hyper-parameters. Convolution and max pool layers are arranged in this manner throughout the entire architecture. After two FC (completely connected) layers, a softmax is used as the output. The 16 in VGG16 stands for having 16 layers and a weight of 16. We used the generative replay technique in our Continual Learning scheme to work our CNN detector model on a series of tasks without being affected by the problem of catastrophe forgetting. The DC GAN Rehearsal method involves storing a portion of the training data from the past and fusing it with fresh patterns for upcoming training. The final technique is called "generative replay," which employs a dual-model approach in which a "current" generative model  $G_t$  is used to learn both the current distribution and the  $G_{t1}$  distribution while a "frozen" generative model  $G_{t1}$  is used to sample from previously learned distributions. A duplicate of  $G_t$  is substituted for the  $G_{t1}$  after a job is finished, allowing learning to continue. We also test three



baselines. A unique generator is trained for each job for Upper bound Data, where the main deepfake architecture is fine-tuned using data from all previous tasks combined.

### 3.4 Algorithm

The algorithm for the proposed GAN-CNN Ensemble is shown in Table 2. It consists of three major sections. The section I shows algorithm about GAN model used to generate new samples using generative replay technique. Section II defines the CNN model used for the detection of deepfakes. And finally, GAN-CNN ensemble model used for implementation of generative replay technique for removing catastrophic forgetting issue of CNN to make a more robust model for deepfake detection.

**Table 2.** Proposed algorithm for the implementation of GAN-CNN Ensemble model.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>(I) GAN (Generative Adversarial Network).</b></p> <p>Generator (G) and Discriminator (D) components of the GAN.</p> <p><math>G(z_i)</math>. Synthetic data generated by the generator G for the noise vector <math>z_i</math>.</p> <p><math>D(x_j)</math>. The output of the discriminator D for the input data <math>x_j</math>, representing the probability that <math>x_j</math> is real (as opposed to fake).</p> <p>The GAN's optimization objective involves the following loss functions.</p> <p>Generator Loss (<math>L_G</math>).</p> $L_G = -\log(D(G(z_i)))$ <p>Discriminator Loss (<math>L_D</math>).</p> $L_D = -\log(D(x_j)) - \log(1 - D(G(z_i)))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>(II) CNN (Convolutional Neural Network).</b></p> <p>The CNN is used for deep fake detection.</p> <p><math>P(x_j)</math>. The predicted probability by the CNN that the input <math>x_j</math> is genuine (real).</p> <p>The CNN's loss function for deep fake detection is.</p> $L_{CNN} = -y_j * \log(P(x_j)) - (1 - y_j) * \log(1 - P(x_j))$ <p>where <math>y_j = 1</math> for genuine images and <math>y_j = 0</math> for deep fake images.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>(III) GAN-CNN Ensemble Model.</b></p> <p>The GAN-CNN Ensemble model combines the GAN's synthetic data and the CNN's predictions.</p> <p>Training Data for CNN.</p> $D_{train} = \{(x_1, y_1), (x_2, y_2), \dots, (x^N, y^N)\} \text{ (N samples)}$ <p><b>Generated Synthetic Data from GAN.</b></p> $D_{synthetic} = \{G(z_1), G(z_2), \dots, G(z_m)\} \text{ (M samples)}$ <p><b>Augmented Training Data.</b></p> $D_{augmented} = D_{train} \cup D_{synthetic} \text{ (N + M samples)}$ <p>The CNN is then trained on the augmented dataset <math>D_{augmented}</math> to learn to distinguish between real and synthetic data while minimizing catastrophic forgetting.</p> <p><b>Ensemble Prediction.</b></p> <p>The ensemble prediction combines the predictions from both the GAN and the CNN.</p> <p>Ensemble Prediction (<math>E_j</math>).</p> $E_j = \text{Ensemble\_Method}(P_{CNN}(x_j), D(G(z_j)))$ <p>where <math>P_{CNN}(x_j)</math> is the output probability from the CNN for input <math>x_j</math>, and <math>D(G(z_j))</math> represents the probability output from the GAN's discriminator for the synthetic data <math>G(z_j)</math>. The ensemble method combines these probabilities to obtain the final decision <math>E_j</math> on deepfake detection for each input <math>x_j</math>.</p> |

## 4. Results and Discussion

The training procedure reveals that the performance of the deepfake detection model improves dramatically as the number of epochs increases. The model improves in correctly detecting genuine and deep fake images, and its predictions match the true labels well, resulting in high accuracy and low loss. Table 3 below displays the performance of a deepfake detection model as it is trained across many epochs. Each row represents the outcomes

gained throughout the training process at a certain epoch. On a validation or test dataset, the model's performance is measured in terms of accuracy and loss. The accuracy is 63.13% and the loss is 0.6482 at the start of training (Epoch 1). This demonstrates that the model's performance is low, and the forecasts are not particularly accurate. Both accuracy and loss begin to increase as training advances. At Epoch 10, the accuracy rises to 88.5%, demonstrating that the model is becoming more adept at distinguishing between genuine and deep fake images. The loss also drops dramatically to 0.286, indicating that the model's predictions are improving.

Table 3. Table showing accuracy and loss values obtained at different epochs.

| EPOCHS | ACCURACY (%) | LOSS   |
|--------|--------------|--------|
| 1      | 63.13        | 0.6482 |
| 10     | 88.5         | 0.286  |
| 20     | 94.4         | 0.157  |
| 30     | 96.44        | 0.1102 |
| 40     | 96.54        | 0.1012 |
| 50     | 98.35        | 0.0572 |
| 60     | 98.83        | 0.04   |
| 70     | 97.42        | 0.0777 |
| 80     | 98.08        | 0.056  |
| 90     | 98.75        | 0.0381 |
| 100    | 98.67        | 0.0337 |

The model's performance improves significantly with more training, with an increase in accuracy and decrease in loss values over time. By Epoch 50, the model's accuracy reached 98.35%, and its loss decreased to 0.0572. It performs well in identifying genuine and fake images, with predictions close to actual labels. The model's performance fluctuates slightly in subsequent epochs, but this is normal when parameters are fine-tuned to fit the training data. At Epoch 100, the model's ultimate performance was 98.67% with a loss of 0.0337, indicating its robustness and ability to generalize well to new data shown in Fig. 5 and 6.

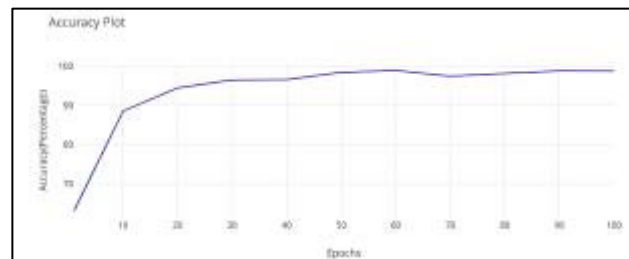


Fig. 5. Accuracy Curve of The Proposed GAN\_CNN Ensemble

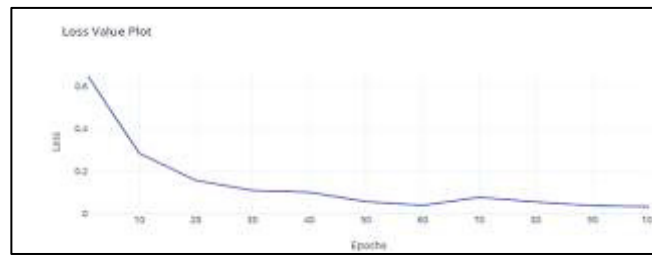


Fig. 6. Loss Curve of the Proposed GAN\_CNN Ensemble.

Table 4 presents the evaluation parameters for a binary classification model for deep fake detection, focusing on two classes (0 and 1). Class 0 and class 1. The model achieved a precision of 0.68 for class 0 and a recall of 0.74 for class



0 items, with a balanced performance. The dataset contains 598 class 1 samples, with a precision of 0.72 and a recall of 0.66 for class 1 items. The model's F1-score is 0.69, indicating a balanced performance.

Table 4. Table showing values of different evaluation parameters (Precision, Recall, F1-score).

| Evaluation Parameters | precision | recall | f1-score | support |
|-----------------------|-----------|--------|----------|---------|
| 0(Fake)               | 0.68      | 0.74   | 0.71     | 598     |
| 1(Real)               | 0.72      | 0.66   | 0.69     | 602     |

Fig. 7 shows the graphical representation of accuracy and recall values for classes 0 and 1, with lower false positive rates indicating higher accuracy and lower false negative rates indicating higher recall. The F1 score measures precision and recall, particularly useful in class imbalances, and the support value indicates sample count.

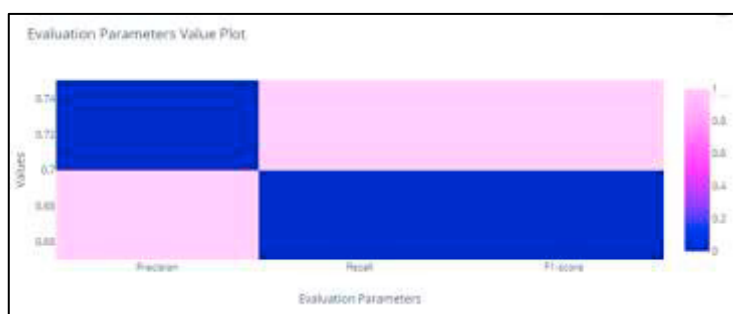


Fig. 7. Evaluation Parameters Curve of the Proposed GAN\_CNN Ensemble

Table 5 shows a confusion matrix assessing a binary classification model's performance by counting true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions. The high TP value indicates a significant detection of deep fake samples, while the high TN score indicates a significant detection of authentic samples. The FP value indicates the number of false positives, indicating the model misclassified 155 actual data as fakes, and the FN value indicates the number of false negatives, misclassifying 204 samples as genuine.

Table 5. Table showing values of True Positive, False Positive, True Negative and False negative of Confusion matrix for proposed model.

| Confusion Matrix   | Actual Positive | Actual Negative |
|--------------------|-----------------|-----------------|
| Predicted Positive | 444             | 155             |
| Predicted Negative | 204             | 398             |

The graphical representation in Fig. 8 provides a more thorough evaluation of the model's performance in binary classification tasks for deep fake detection. Furthermore, the confusion matrix shows that the model properly recognizes a significant proportion of real and deep fake samples (True Positives and True Negatives). However, there are certain cases of misclassification (False Positives and False Negatives), indicating that the model needs to be improved further.

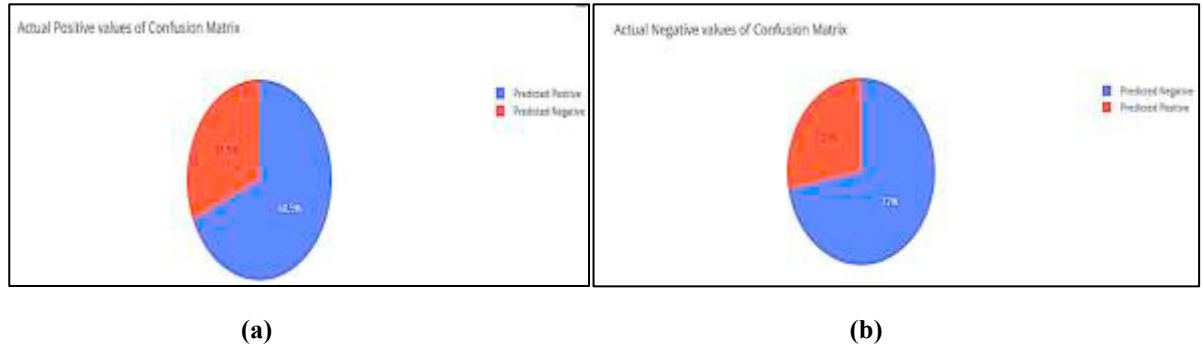


Fig. 8. Confusion Matrix with Actual Positives (a); and Actual Negatives(b) of the Proposed model

The GAN model's training progress is shown in table 6, demonstrating its effectiveness in reducing catastrophic forgetting issues encountered by CNN models in deepfake detection. The model's performance is reflected in data, including epochs, time required, generator loss, and discriminator loss. The generator loss decreases from 2.5187 at epoch 100 to 0.0777452, indicating an improvement in the generator's ability to generate data like real datasets.

Table 6. Performance of Generator and Discriminator modules used in GAN-CNN Ensemble

| Epochs | Time(sec) | Generator Loss | Discriminator Loss |
|--------|-----------|----------------|--------------------|
| 0      | 0.086946  | 2.5187         | 0.09220361         |
| 10     | 0.082465  | 3.2298         | 0.839503           |
| 20     | 0.0761921 | 3.9640         | 0.02143725         |
| 30     | 0.0961961 | 4.5883         | 0.014482           |
| 40     | 0.1040106 | 4.9379         | 0.0160632          |
| 50     | 0.0943568 | 5.2372         | 0.007435           |
| 60     | 0.0836747 | 5.5832         | 0.005152           |
| 70     | 0.1040618 | 5.9138         | 0.004010           |
| 80     | 0.0726762 | 5.9825         | 0.0027085          |
| 90     | 0.0832026 | 6.10523        | 0.00272559         |
| 100    | 0.0777452 | 6.0917         | 0.00233806         |

The discriminator loss decreases from 0.09220361 to 0.00233806, indicating better discrimination of genuine data from produced data. This decrease in loss values for both the generator and discriminator indicates successful training. As the generator loss decreases and the model improves at creating data, the discriminator loss decreases as well, making it harder for the discriminator to distinguish between actual and produced data. The model's success is measured by the balance between generator and discriminator losses and how well it generalizes to fresh data after training shown in Fig. 9 and 10 below.

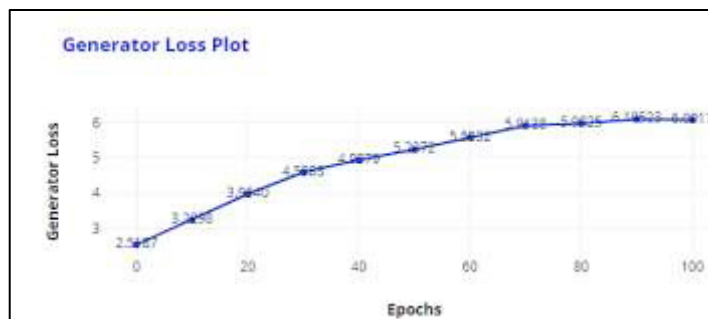


Fig. 9. Generator Loss Plot of the Proposed GAN-CNN Ensemble model

The GAN model's generator and discriminator loss begin at 2.5187 and 0.09220361, respectively, at the start of training. As training progresses, the generator loss decreases, indicating the model's ability to generate synthetic data that closely resembles real data. By Epoch 100, the generator loss drops to 0.0777452, while the discriminator loss decreases to 0.00233806, indicating the discriminator's difficulty in distinguishing between actual and created data. These dynamic highlights the GAN's effective adversarial learning process, where the generator and discriminator continuously improve their performance.

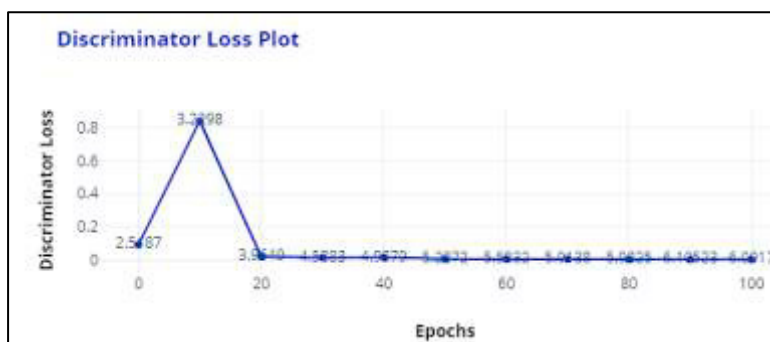


Fig. 10. Discriminator Loss Plot of the Proposed GAN-CNN Ensemble model

Fig. 11 shows the model's time complexity during training. The first 30 epochs have stable time, but a spike around the 40th indicates a temporary surge in processing. Despite this, the model maintains efficient time management throughout 100 epochs.

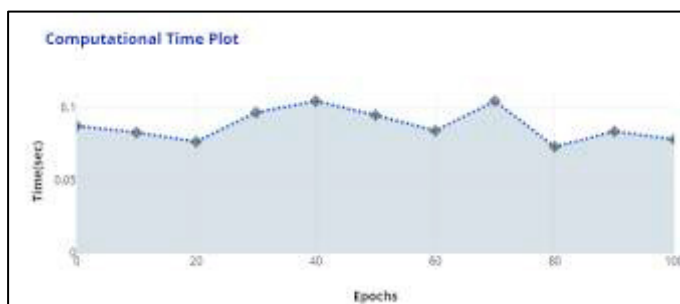


Fig. 11. Generator Loss Plot of the Proposed GAN-CNN Ensemble model

Time management is critical for a model's effectiveness, particularly in deepfake detection tasks. It enables the model to efficiently analyze and respond to data, especially during volatility. Balancing time efficiency and learning efficacy is critical for effective implementation in scenarios that need rapid and accurate decision-making. This performance reflects the model's ability to overcome catastrophic forgetting, optimize deepfake detection, and balance efficient time management with adversarial learning. GAN-CNN Ensemble model is found to have a significant potential for deepfake detection, with high accuracy and effective handling of catastrophic forgetting in comparison with existing models as shown in Table 7 below. However, further analysis and fine-tuning may be required to minimize misclassifications and optimize the model's performance in real-world circumstances.

In comparison to previous techniques for detecting real and fake faces, our GAN-CNN Ensemble model exceeds the competition with an astonishing 98.67% accuracy. This outperforms Afchar et al.'s Face Forensics++ Voting Ensemble (94.05%), Nugen et al.'s Stacking Ensemble in the Deepfake Detection Challenge (DFDC) (97.97%), Yang et al.'s AdaBoost Ensemble in DFDC (97.30%), Li et al.'s UADFV Voting Ensemble (98.03%) as well as Wang et al.'s DFDC Ensemble with Voting (96.40%). The GAN-CNN Ensemble model outperforms deepfake detection with high accuracy, demonstrating its robustness in this challenging context. These results demonstrate the model's effectiveness in pushing the boundaries of face detection, making it a significant advancement. Our approach's

superior performance in deepfake detection is a promising solution for real-world applications, offering a strong foundation for future research to explore its potential across diverse datasets and assess its adaptability to changing data distributions.

Table 7. Comparison of the Proposed Model with Existing Models

| Author             | Dataset                      | Approach          | Result |
|--------------------|------------------------------|-------------------|--------|
| Ours               | Real and Fake Face Detection | GAN-CNN           | 98.67% |
| Afchar et al. [20] | FaceForensics++              | Voting Ensemble   | 94.05% |
| Nugen et al. [21]  | DeepFake Detection Challenge | Stacking Ensemble | 97.97% |
| Yang et al. [22]   | DFDC                         | AdaBoost          | 97.30% |
|                    |                              | Ensemble          |        |
| Li et al. [23]     | UADFV                        | Voting Ensemble   | 98.03% |
| Wang et al. [24]   | DFDC                         | Voting Ensemble   | 96.40% |

In a nutshell, the proposed GAN-CNN Ensemble model has shown significant improvements in deepfake detection performance, with an impressive accuracy rate of 98.67% after 100 epochs. The model's ability to minimize errors and improve precision has also improved. Assessment parameters show significant values for precision and recall, with F1-scores reaching 0.71 for the 'Fake' class and 0.69 for the 'Real' class. The model consistently manages its computing time during training, ensuring practical applicability. The effectiveness of the model depends on a compromise between efficient time management and enhancing accuracy and loss values. It outperforms various alternatives in comparison to existing methods, providing solid evidence for its usefulness in deepfake identification. Key improvements include increased accuracy, robustness, effective time management, and improved performance, making it an important contribution to the field.

## 5. Conclusion

The research successfully addressed the CNN detector's catastrophic forgetting issue in deepfake detection for social media images by implementing a Generative Replay technique using a GAN-CNN ensemble model. This method includes generating and saving samples from previous tasks, then replaying them during new task training, considerably improving CNN's robustness in detecting deepfakes. It has been observed that the ensemble model exhibited excellent performance by achieving an outstanding accuracy of 98.67% (training data), and 70.08% (testing data) with low loss (0.0337). Furthermore, for classes 0 and 1, it displayed strong precision values of 68% and 72%, recall values of 74% and 66%, and F1 scores of 71% and 69%, respectively. These findings emphasize the model's ability to manage catastrophic forgetting and its effectiveness under dynamic training conditions. The future scope involves evaluating the model's performance on diverse datasets and ensuring its adaptability to changing data distributions and potential adversarial threats in real-world applications.

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